CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

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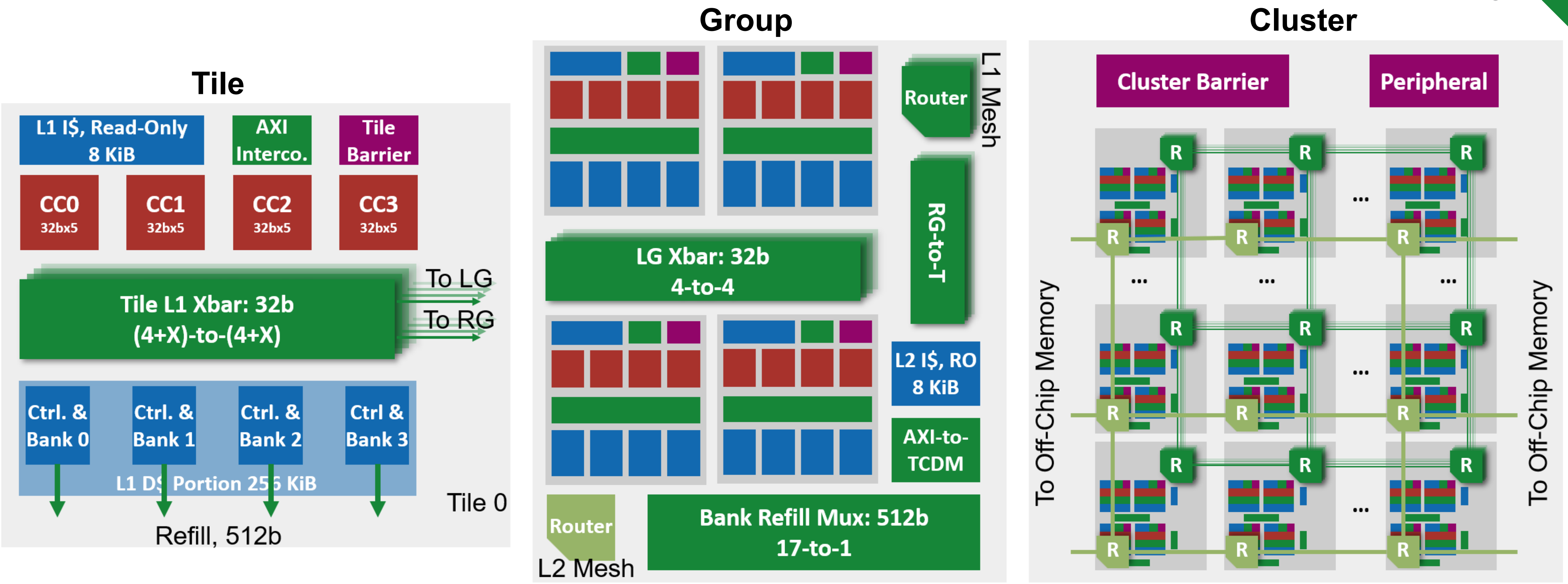
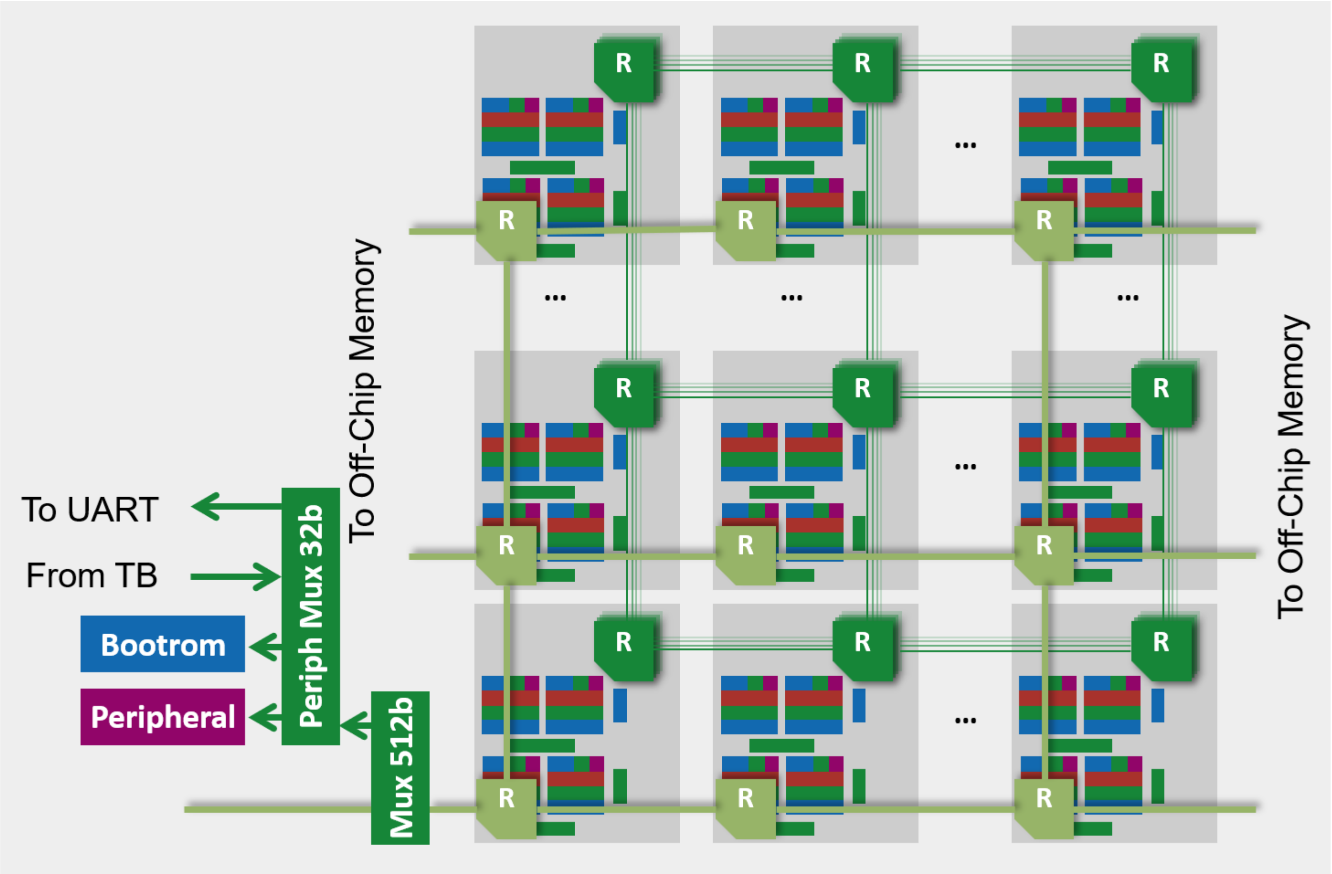
Hardware Development (Cluster)



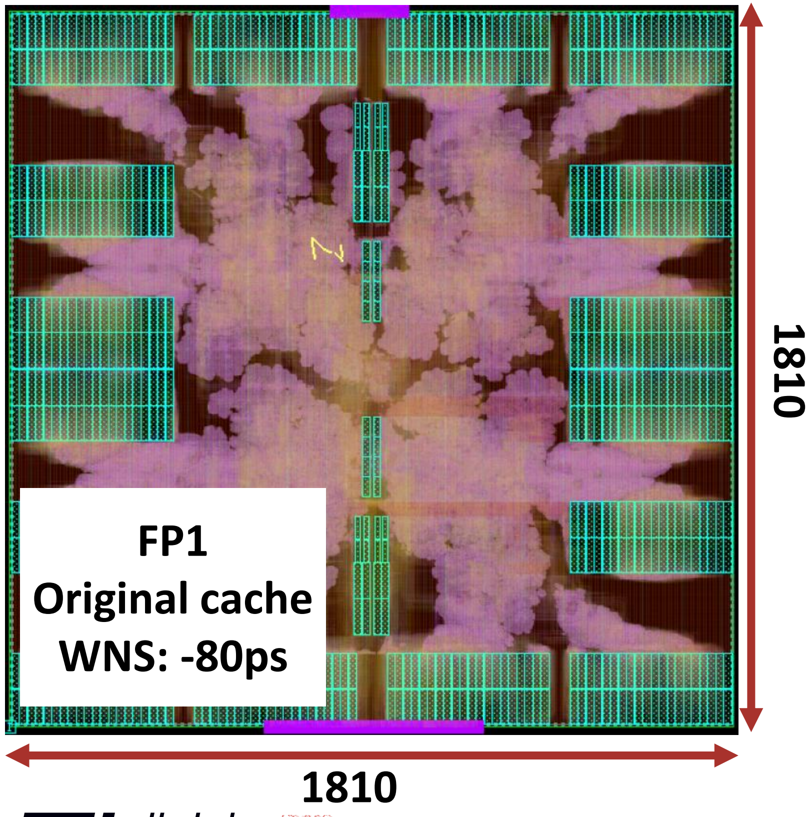
Hardware Development (Cluster)

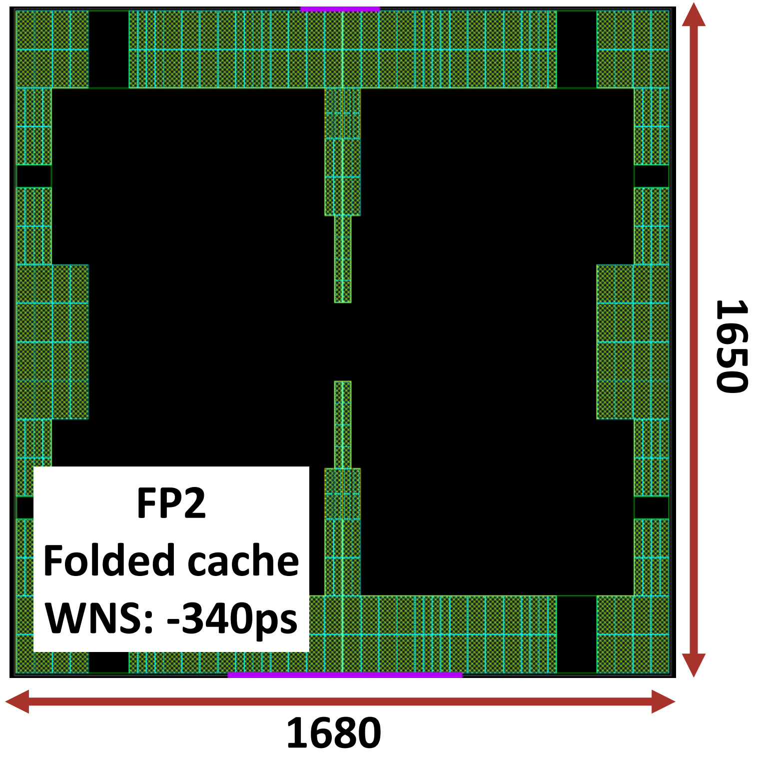
- **Refill NoC**
  - Switch the L2 Refill NoC to TCDM-based
    - Add a tcdm chimney for source-based routing algorithm
    - Put bootrom, peripheral and UART under NoC exit port for DRAM/HBM0
    - Remove the no-longer needed AXI xbars inside tile level for peripheral/bootrom selection
  - Now under functional and performance testing

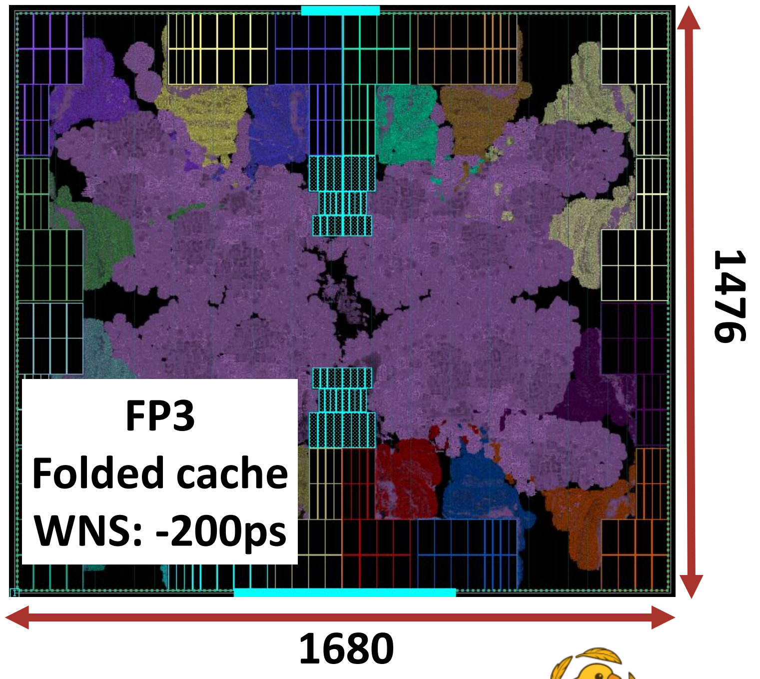


Physical Design (Folded Cache)

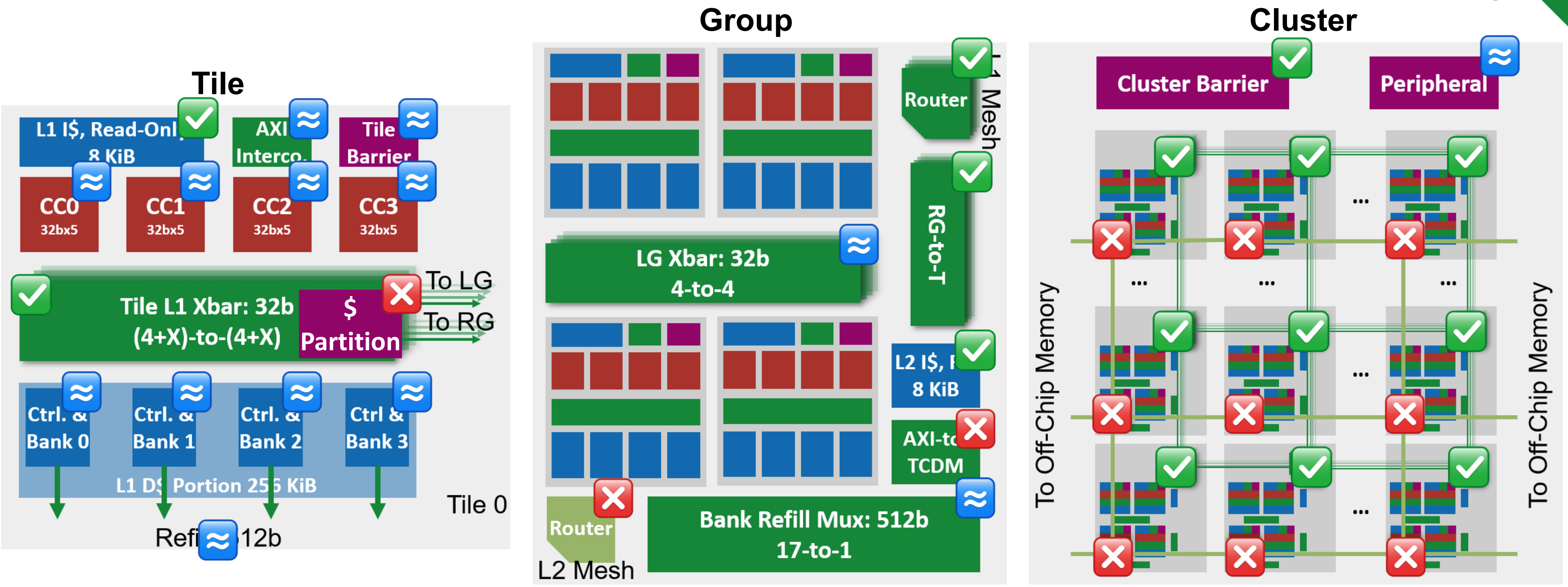
- **Timing**
  - Combine FP3 with the optimized RTL => bring WNS down to **-180ps**
  - Now working on further optimization in RTL level, but at lower priority due to other tasks







GVSoC Modeling Progress



- ✅ Done, the same logic as RTL
- ≈ In place, with approximate model
- ❌ Not implemented yet

GVSoC Modeling Progress

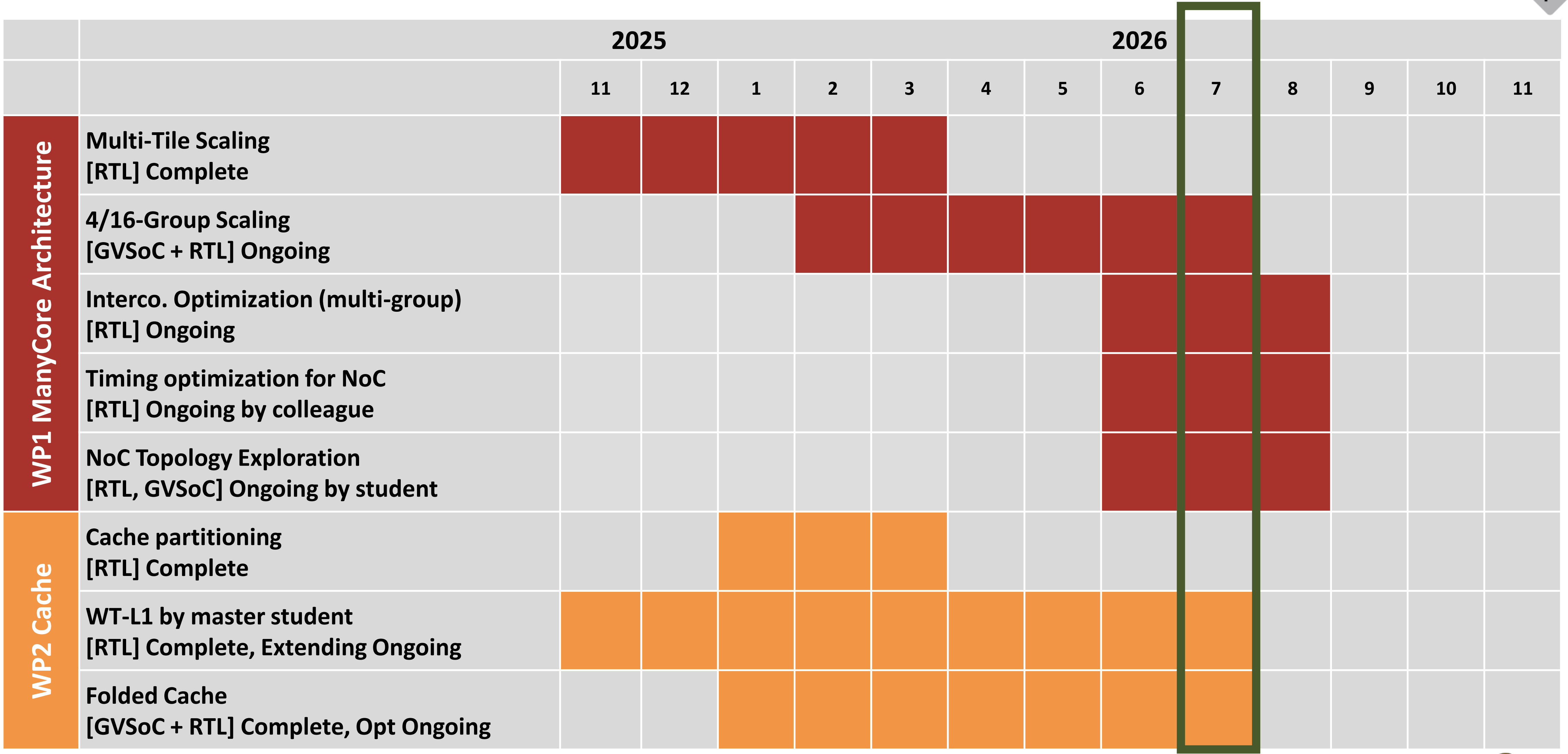
- **Core-Level Modeling**
  - Update to latest Spatz model to support async transactions

- **Cluster-Level Modeling**
  - Extend the model to full cluster scale (256 cores in 16 groups)
  - L2 NoC not yet implement, using the previous Xbar-based approach
    - Align with the stable RTL version
  - No calibration done so far on the model => mismatch can be large at this moment
  - Address scrambling and cache partitioning features are not supported at this moment
  - Passed dotp, matmul, gemv kernels (correct results, lower performance than RTL due to the missing features and calibration)

Next Step

- **Diyou Side**
  - Physical design exploration for multi-group cluster => help to determine the BW we can have
    - NoC router layout and port placements
    - Cluster-level floorplan
  - Multi-Snitch cores sharing one Spatz
    - Two approaches
      1) add heterogeneous CC into tile, containing only Snitch cores
      2) add additional Snitch into the current CC (what we have done in a different project)
    - Some needed additions
      1) Snitch inside the tile should all connect to the Spatz, using muxing to select the host
      2) Software and runtime support to reduce the coding complexity (some helper functions)
- **Zexin Side**
  - GVSoC development and calibration
  - SW development for multi-user

Timeline



2025
2026
11
12
1
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7
8
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10
11
WP1 ManyCore Architecture
Multi-Tile Scaling
[RTL] Complete
4/16-Group Scaling
[GVSoC + RTL] Ongoing
Interco. Optimization (multi-group)
[RTL] Ongoing
Timing optimization for NoC
[RTL] Ongoing by colleague
NoC Topology Exploration
[RTL, GVSoC] Ongoing by student
WP2 Cache
Cache partitioning
[RTL] Complete
WT-L1 by master student
[RTL] Complete, Extending Ongoing
Folded Cache
[GVSoC + RTL] Complete, Opt Ongoing

Timeline

		2025		2026										
		11	12	1	2	3	4	5	6	7	8	9	10	11
WP3 VPE	Shared vector core [RTL + GVSoC] Ongoing													
WP4 Benchmark	Single-user kernel optimization [SW] Ongoing													
	Multi-user and more scenarios [SW]													
Report	Final Report and Benchmark													

L1 Access Latency

- **Problem: increased L1 access latency when scaling up**
- **Solutions**
  - **(Ongoing, RTL & Physical) Hardware optimization**
    Further optimize the interconnection and cache controller to reduce access latency on existing architecture
    => Retiming the current interconnection, remove unnecessary cuts in interco and cache
    => Target to reduce 1-2 cycles (~5 cyc) for hit-to-use latency
    Currently has reduced to 6 cycles:
    Core => Xbar => AMO => Ctrl (2) => AMO => Core
  - **(Complete, RTL) Cache partitioning**
    Partition the shared L1 to keep data localized to avoid remote access latency
  - **(Complete with some problems, RTL & Physical) WT-L1**
    Explore and evaluate a private WT L1 for each core
    => Performance degrade on single tile
    => Needs more work to support larger system

L1 Access Latency

- **Problem: large NoC latency (~28 cyc based on previous exploration)**
- **Solutions**
  - **(Ongoing, Physical) Reduce hop latency**
    Explore the timing under 7nm technology to see the possibility of reducing per-hop latency to 1 cycle
    => Reduce the latency by 1/2
  - **(Planned, Physical) Cut clk-domain (optional)**
    If 1-cyc is not possible, explore to operate the NoC under a higher frequency than cores (e.g. 1.5GHz)
    => Effectively cut the latency by 1/3
  - **(Planned, GVSoC & Physical) NoC Topology**
    Explore different NoC topologies
    => Torus to reduce the diameter of NoC by 1/2, reduce longest latency by 1/2

PE-Related

- **Problem: Improve scalar core performance**
- **Solutions**
  - **(Planned, RTL & GVSoC) Shared vector core**
    Explore two Snitch sharing one Spatz configuration to improve scalar performance
    Software complexity needs to be considered, especially on the possible conflicting use of register file.
    Discussed Solution: critical section

RLC Workload

- **Problem: Current RLC model cannot reveal all properties of the kernel**
- **Solutions**
  - **(Ongoing, SW) Kernel optimization**
    Adapt the current model to better represent the real cases, e.g. do some operations on the data. Further cooperation with HQ side to develop a more complete and better model.
  - **(Planned, SW) Kernel development**
    Implement the missing use cases for multi-user

Thank you!

Q&A